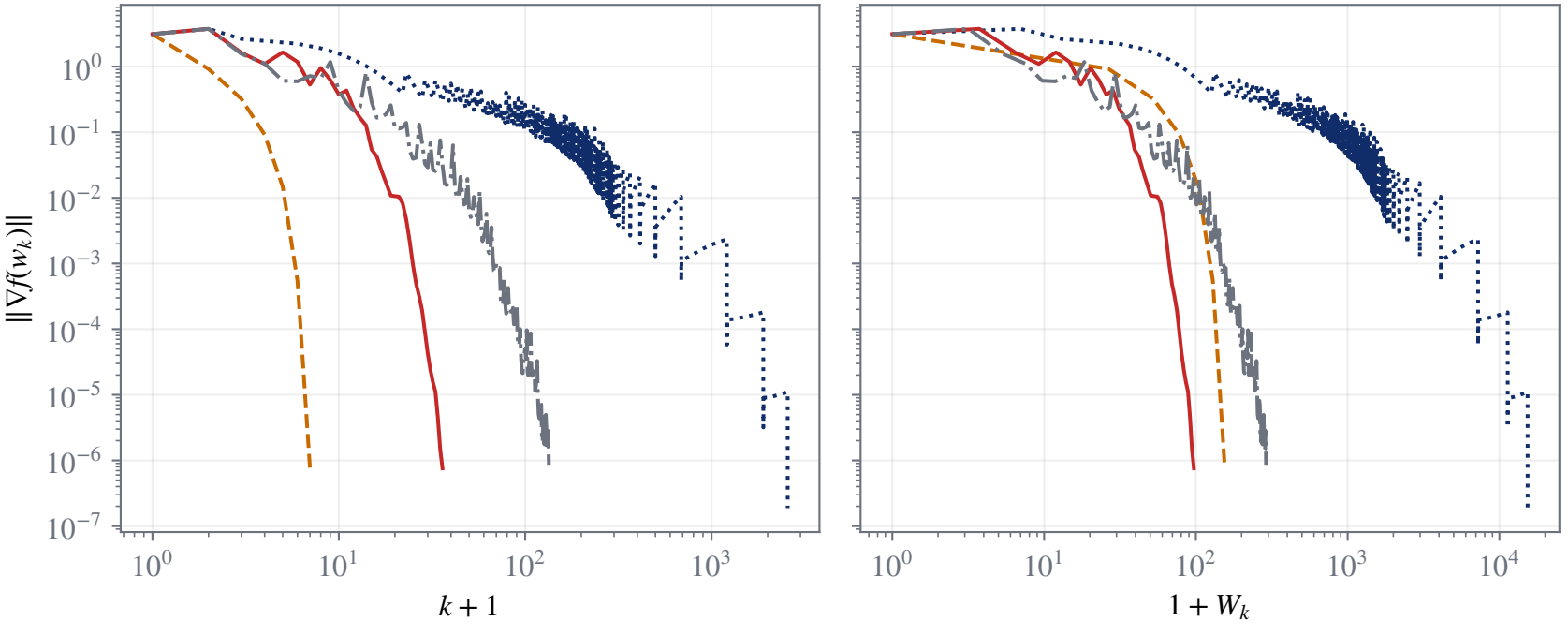
$\|\nabla f(w_k)\|$
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{-7}
10^{0}
10^{1}
10^{2}
10^{3}
$k+1$
10^{0}
10^{1}
10^{2}
10^{3}
10^{4}
$1+W_k$